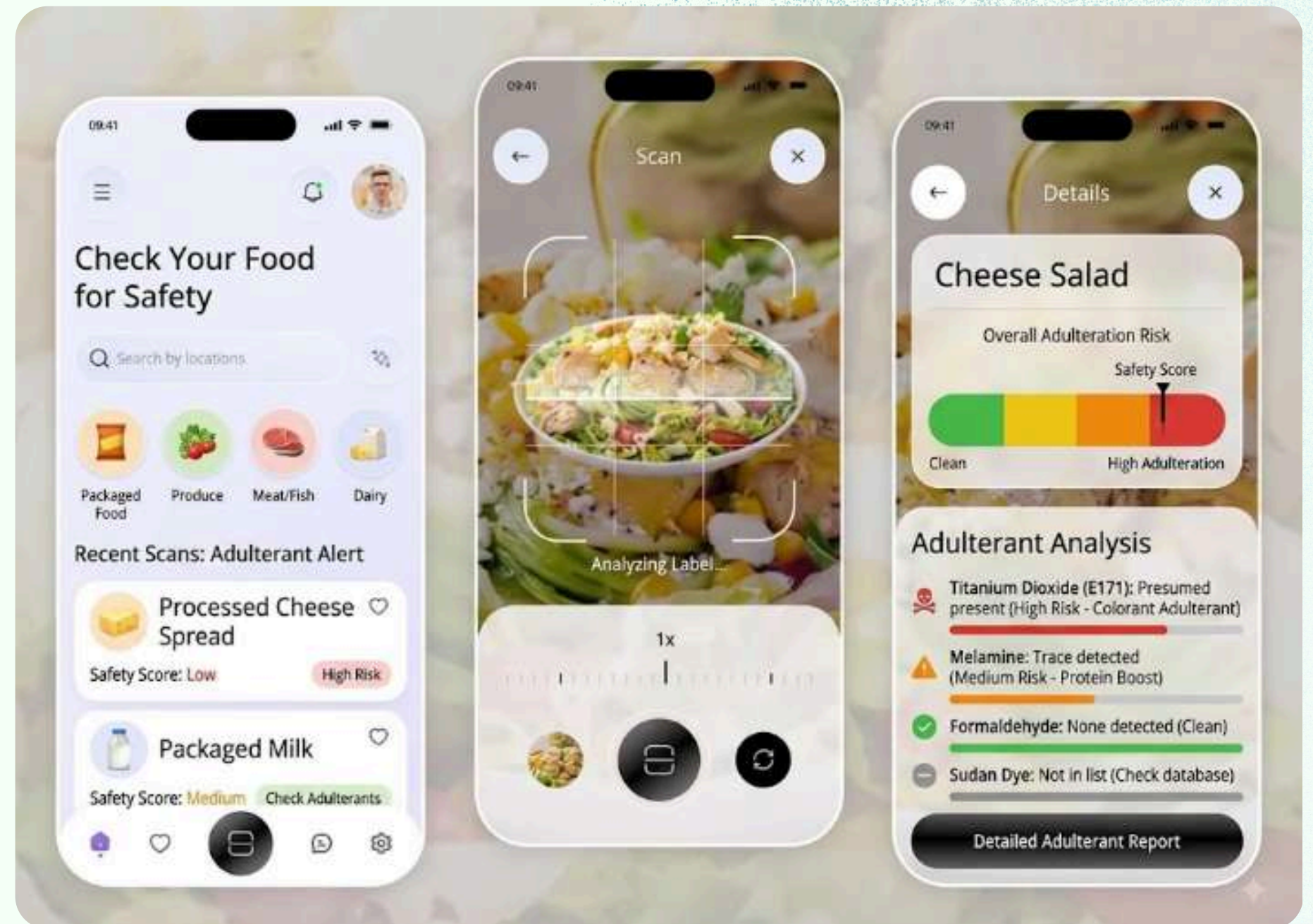


Willow

~Know what's really in it.



AN AI-BASED PRODUCT INGREDIENT SCANNER FOR THE INDIAN MARKET

Introduction

What the application does?

A user photographs a packaged product. The app reads the ingredient list from the image, flags any ingredients known to be harmful, explains their health effects in plain language, and suggests healthier alternatives, all without relying on a barcode database.

FUNFACT:- While India uses mandatory Green/Red dots for Veg and Non-Veg items, these apps expose hidden non-vegetarian additives (like gelatin, bone-char processed sugar, or E120 carmine dye) that might otherwise go unnoticed.

Flow-of-project



Photo, not barcode

WORKS ON ANY PRODUCT – NOT JUST ONES IN A BARCODE DATABASE.



Plain-language flags

HARMFUL INGREDIENTS EXPLAINED IN TERMS ANYONE CAN UNDERSTAND.



Healthier alternatives

COMPARABLE PRODUCTS SUGGESTED IN THE SAME PRICE RANGE.

Motivation

Why this, why now

Analysis

China rejects Indian chillis, Japan our mangoes. Who is checking what we eat?

None of this is new, and that's the point.

By: Shardool Katyayan 19 Jun, 2026

60,000 Kg

of adulterated spices seized
in Gujarat in a single sweep
(Apr 2024)

2

major Indian spice brands
banned abroad after
pesticide contamination
found

0

solutions existing tuned
according to indian market
right now

Rising health awareness meets an information gap: Indian consumers increasingly want to know what they're eating and applying, but existing scanning apps are built around Western labeling standards and barcode-indexed brands — leaving most of the Indian market unscanned.

Three gaps existing tools don't cover

- **BARCODE-FIRST TOOLS MISS LOCAL PRODUCTS**

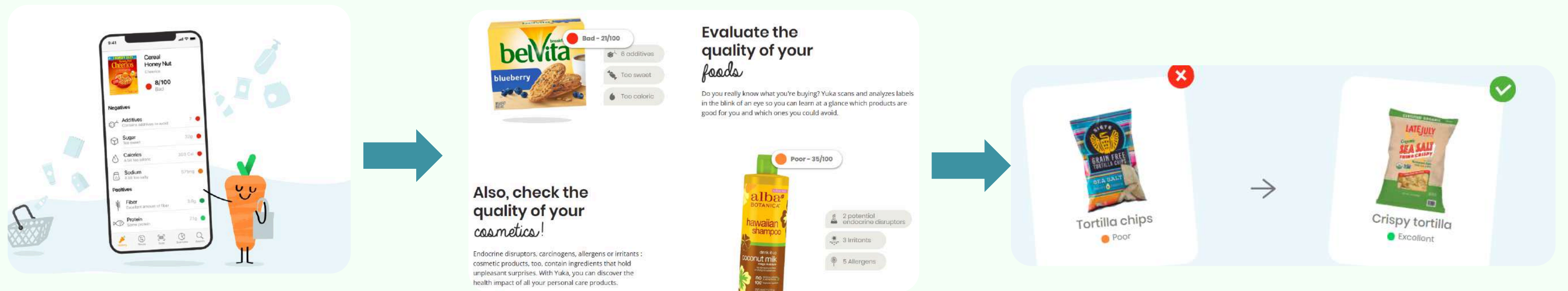
Global apps like Yuka rely on barcode databases — unbranded and regional Indian products are rarely indexed, so most local packaged food goes unscanned.

- **NO PRINTED INGREDIENT LIST AT ALL**

Local sweets, bakery items, and loose-packaged snacks are common in India and often carry no ingredient panel to read in the first place.

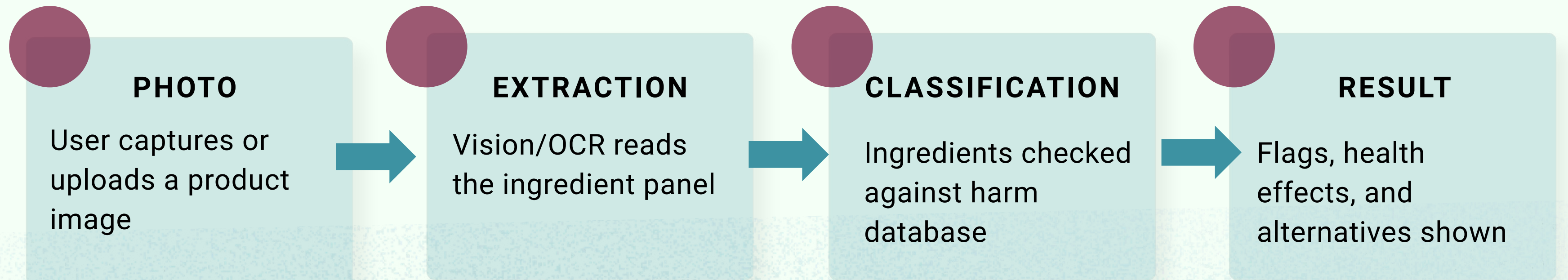
- **STANDARDS BUILT FOR A DIFFERENT MARKET**

Existing scoring is calibrated to EU/US additive standards, not FSSAI guidance, Indian allergens, or Ayurvedic ingredients.



Market-Centric

"YUKA SCANS BARCODES FOR GLOBAL BRANDS; WILLOW SCANS PACKAGING PHOTOS TO BRING FOOD SAFETY TRANSPARENCY TO EVERY LOCAL AISLE IN INDIA."



FOR PRODUCTS WITH NO READABLE INGREDIENT LIST , **A COMMON CASE FOR LOCAL INDIAN BRANDS** — THE APP FALLS BACK TO A GENERIC, CATEGORY-LEVEL RISK PROFILE INSTEAD OF GUESSING, SO IT STAYS USEFUL AND HONEST.

Built for the Indian market specifically

IMAGE-FIRST, NOT BARCODE-FIRST

Reads any product's ingredient panel directly — covers local and unbranded brands most scanners miss.

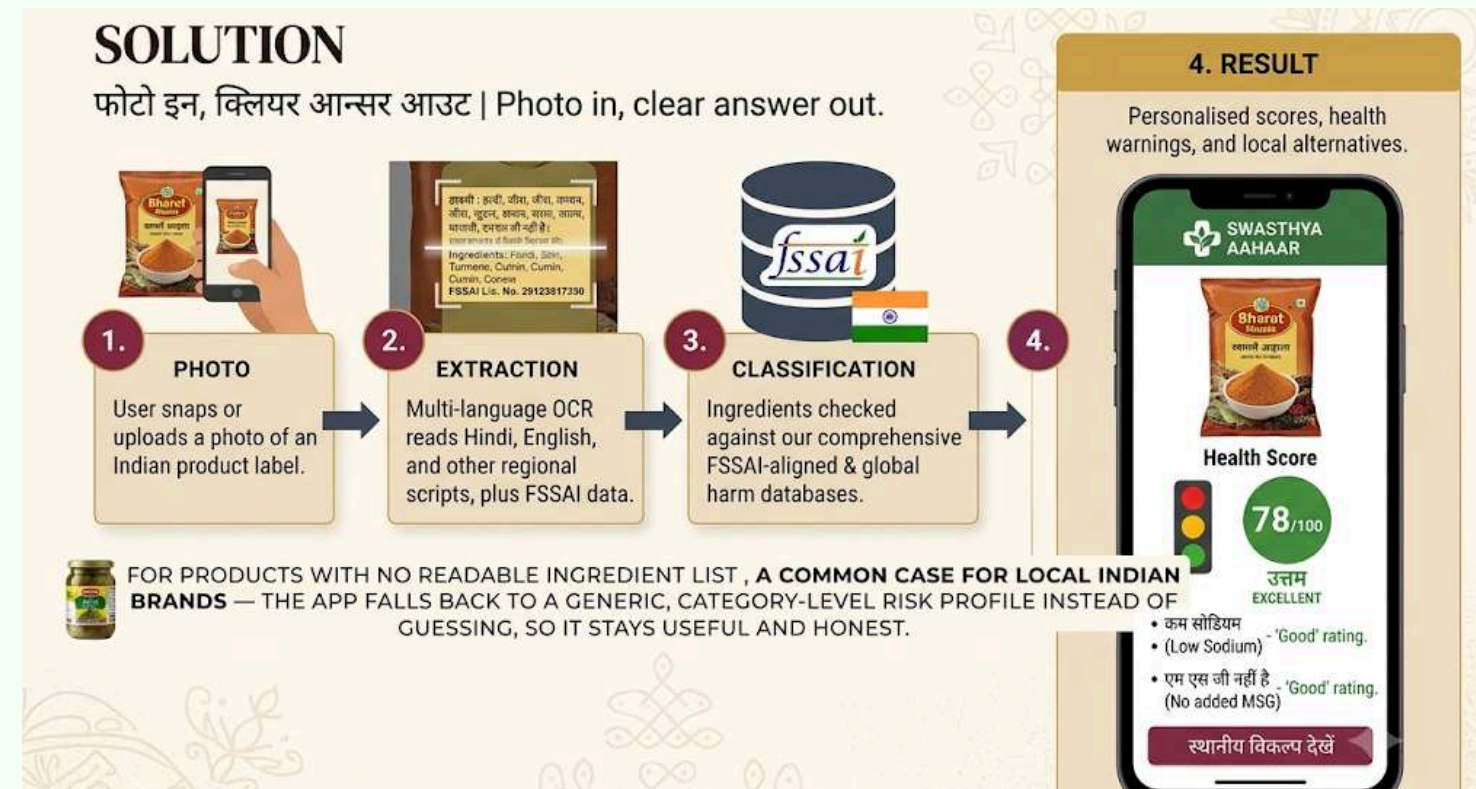
CATEGORY-BASED FALLBACK

No printed list? The app estimates risk from product category instead of failing silently.

BUILT TO EXTEND

New categories (cosmetics, spices) and languages plug into the same architecture — no rewrite needed.

- Vision-Based
- Fallback-Ready
- Normalized
- Suggestion of better products



- Budget-Matched
- Extensible
- Traceable
- Voice-Accessibility (TTS)

What's our next step?

Standardize & Merge Open Datasets

- Open Food Facts Global Data Dump

link ~ <https://world.openfoodfacts.org/data>

- Indian Packaged Foods Nutritional Dataset

link ~ <https://world.openfoodfacts.org/data>

- Food Adulteration & Risk Severity Dataset

link ~ <https://www.kaggle.com/datasets/fahmidachowdhury/food-adulteration-dataset>



Thank You~